BENNETT UNIVERSITY
THE TIMES GROUP

Enrolment No: ____________________ Name of Student: ____________________

Department/ School: ____

END-TERM MAKE-UP EXAMINATION, ODD SEMESTER DECEMBER JAN 2024

COURSE CODE	**CSET-203**	**MAX. DURATION**	**2HRS**
COURSE NAME	**MICROPROCESSOR AND COMPUTER ARCHITECTURE**		
PROGRAM	**B.Tech.**	**TOTAL MARKS**	**35**

Mapping of Questions to Course and Program Outcomes										
Q.No.	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5
CO	1	1	2	1	2	1	2	2	3	3
PO	1	1	1, 6	1	1, 6	1, 3	6	1	1, 6	1, 6
BTL*[1]	L1	L1	L2	L1	L3	L1	L5	L4	L4	L5

GENERAL INSTRUCTIONS: -

1. Do not write anything on the question paper except **name, enrolment number** and **department/school.**
2. Carrying mobile phones, smartwatches and any other non-permissible materials in the examination hall is an act of **UFM.**

COURSE INSTRUCTIONS:

a) **Be Precise in your writing**

SECTION A

Max Marks: 10

A1) Given multiplicand m=0101 1010 1110 1110 and multiplier n=0011 0101 1001 1101. Product P =m x n is obtained by multiplication of two numbers using the Booth's algorithm. Find out the number of total additions and subtractions performed to find the product. **(2 Marks)**

A2) List the four differences between Hard disk drive (HDD) and solid-state drive (SDD). **(2 Marks)**

A3) A machine has two way set associative cache with following characteristics. Main memory of 4 MB, word size of 1 byte, block size of 64 words, cache size of 8 KB. Find the number of SET and TAG bits. **(1+1= 2 Marks)**

A4) Differentiate between RISC and CISC processors in context with high end processors. **(2 Marks)**

A5) Perform the addition **-5+7** when the two numbers involved are written in Two's complement representation: **(2 Marks)**

SECTION B

Max Marks: 25

B1) Explain with block diagram the working of DMA mode of data transfer in processors. **(5 Marks)**

B2) Use the Booth's algorithm to multiply 23 (multiplicand) by 29 (multiplier), where each number is represented using 6 bits. **(5 Marks)**

B3) Suppose a DMA controller transfer 6-bit word memory using cycle stealing. The device that transmits characters at a rate of 1200 characters per second. The CPU average transfer rate is of 1 million instructions per second. Compute the percentage with which the processor will be slow down due to DMA activity. **(5 Marks)**

B4) We are given with 16K x 4 RAM Chips. Calculate the number of chips required to build 64KB RAM. Also, calculate the size of the decoder if used. Draw the final designed circuit as well **(2+1+2 = 5 Marks)**

B5) Imagine a pipeline system that has four stages: instruction fetch (IF), instruction decodes (ID), execution (EX) and write back (WB). The following table illustrates the number of clock cycles needed for each instruction to finish each stage. Calculate the number of clock cycles and stalls required to complete the instructions. **(5 Marks)**

Instructions	Instruction Fetch (IF)	Instruction Decode (ID)	Execution (EX)	Write Back (WB)
I_0	2	1	2	1
I_1	2	2	3	1
I_2	1	2	1	2
I_3	2	3	2	1
I_4	3	1	1	2

-ALL THE BEST-